

Radiology & Medical Imaging: The Photonic Frontier

Seeing through the smoke
of symptoms. Stay fluffy!



KayfaBIZARRO!
Dive in, nerds!
Stay fluffy!



Roentgen-Ray (The Classic X-Ray)

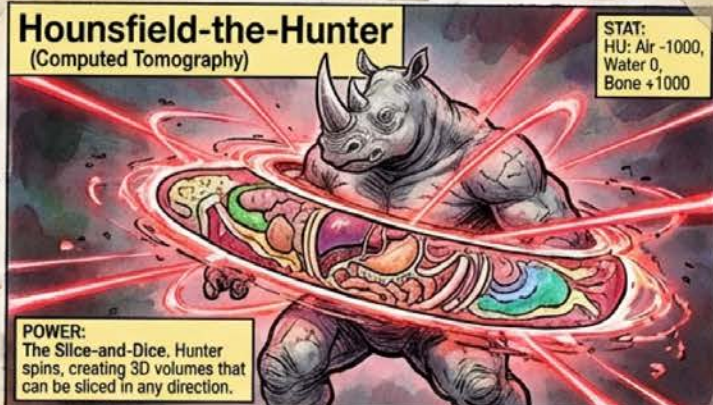


STAT:
Ionizing radiation
2D Projection
Summation effect.

POWER:
The Density-Mapper. Ray shoots through the body; bones stop him (White), air lets him fly (Black), everything else is gray.

LORE: Calcium in bones is a Heavy-Hitter that blocks the beam, creating the shadow-silhouette.

Hounsfield-the-Hunter (Computed Tomography)



STAT:
HU: Air -1000,
Water 0,
Bone +1000

POWER:
The Slice-and-Dice. Hunter spins, creating 3D volumes that can be sliced in any direction.

LORE: X-Ray-Attenuators calculate precise density. Distinguishes Fresh-Blood (+60 HU) from Simple-Fluid (0 HU).

KayfaBINGO! Watch the magnets!

Echo-the-Sonar (Ultrasound)



STAT:
High-frequency sound
No radiation
Piezoelectric

POWER: The Bounce-Back. Fluid is Black-Silent (Anechoic); Bone is White-Noisy (Hyperechoic).

LORE: Based on Acoustic Impedance. Watches the Heart-Pump in real-time without hurting it.

Mag-the-Magnet (MRI)



STAT:
Tesla Strength (1.5T / 3T)
T1 & T2 relaxation

POWER: The Proton-Spinner. Mag makes the body's Water-Protons dance to his magnetic rhythm.

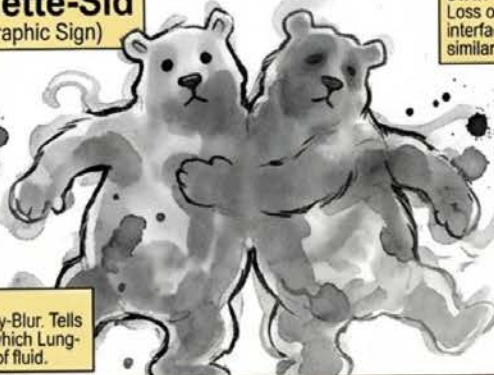
LORE: Unlocks Soft-Tissue-Secrets. Sees the nerves and ligaments that Roentgen-Ray completely misses.

Silhouette-Sid (Radiographic Sign)

STAT:
Loss of normal
interface between
similar densities.

POWER:
The Boundary-Blur. Tells
you exactly which Lung-
Lobby is full of fluid.

LORE: You only see a border when Air meets Water. If Water meets Water, the line is erased.



Kerley-the-Lines (Pulmonary Edema)

STAT:
Kerley B lines /
Short horizontal
peripheral lines.

POWER:
The Fluid-Ladder. When the
Heart-Pump fails, Lymphatic-Orains
overflow, creating white ladders.

LORE: Thickened interlobular septa. The first sign the lungs are becoming a Swamp.

KayfaBINGO! Watch the magnets!

Air-Broncho-Bill (Consolidation)

STAT:
Air-filled bronchi
outlined by fluid-filled alveoli.

POWER: The Tubular-Glow. Shows the Air-Pipes running through a solid Infection-Block.
LORE: Proof that Alveolar-Balloons are full of pus, but Airway-Tunnels remain open.

Diagnosis?! BLÖDSINN!



Wedge-the-Infarct (Pulmonary Embolism)

STAT:
Hampton's Hump /
Peripheral wedge
opacity.

POWER: The Dead-Zone. The PE-Phantom blocks an artery, a wedge of tissue dies and turns white.
LORE: Ischemic necrosis of parenchyma. Revealed by a Filling-Defect in the pipe on CT-Angio.



Rigler-the-Double-Wall (Pneumoperitoneum)

STAT:
Both sides of the
bowel wall are
visible.

POWER:
The Leak-Detector. Appears
when a pipe pops, letting
air escape into the belly.

LORE: Normally you only see the inside. If you see the outside too, there is Free-Air-Frenzy.

Sentinel-the-Loop (Local Ileus)

STAT:
Single dilated loop
of small bowel near
inflammation.



POWER: The Crime Scene-Pointer. Stays perfectly still and bloated, pointing at the Inflammation-Riot.

LORE: Localized paralysis of the gut caused by nearby Cytokine-Chaos.

Coffee-Bean-Carl (Sigmoid Volvulus)

STAT:
Massive dilated
colon loop shaped
like a coffee bean.

POWER:
The Twist-Indicator. Shows
exactly where the colon has
twisted on its Blood-Supply-Anchor.

LORE: A closed-loop surgical emergency. If not untwisted, it becomes Necrosis-Nick.

KayfaBINGO! Watch the magnets!

String-Sign-Steve (Crohn's Disease)

STAT:
Narrowing of terminal
ileum on Barium Swallow.



POWER: The Stricture-Signal. Shows where chronic inflammation turns the Gut-Highway into a Narrow-Thread.

LORE: Transmural scarring and spasm. If the thread completely breaks, a Fistula forms.

Midline-Mitch (Mass Effect)

STAT: Shift of septum pellucidum > 5mm.

POWER: The Brain-Bully. Occurs when a Brain-Bleed pushes the whole brain-center to one side.

LORE: The skull is a Fixed-Box. If Mitch pushes too hard, he causes Herniation-Harry—the total system crash.



Crescent-Chris (Subdural Hematoma)

STAT: Concave, crescent shape / Crosses sutures.

POWER: The Slow-Squeeze. A Venous-Leak that slowly wraps around the brain like a creeping shadow.

LORE: Tearing of Bridging-Veins. Common in Elderly-Wrestlers after a minor Tumble.



Lemon-Lenny (Epidural Hematoma)

STAT: Convex, lens shape Does NOT cross sutures.

POWER: The High-Pressure-Punch. An Arterial-Burst building fast, punching a sharp lens-shape.

LORE: Tearing of Middle Meningeal Artery. Features a Lucid-Interval before total blackout.



KayfaBOGGLE!
Watch the midline!

FLAIR-the-Light (MRI Sequence)

STAT: Fluid Attenuated Inversion Recovery.

POWER: The Edema-Finder. Turns off the Water-Noise of normal CSF so only Inflammation-Fluid glows.

LORE: Essential for spotting MS-the-Plaque-Maker or early Stroke-Steve.



GADOLINIUM-GARRY (MRI Contrast)

STAT: Paramagnetic
Shortens T1 relaxation.



POWER: THE BARRIER-BREAKER.
Only shows up where the Blood-Brain-Barrier is broken by Tumor-Max.

LORE: Garry makes things Bright-White on T1. If it glows, it's growing or it's angry.

BARIUM-THE-BULWARK (Oral/Rectal Contrast)

STAT: Inert chalky suspension.



POWER: THE GUT-COATER.
Barium paints the Digestive-Tunnel lining, revealing Percy-the-Ulcer.

LORE: High density allows for Double-Contrast (Air + Barium) to see incredibly fine surface detail.

IODINE-THE-ION (CT Contrast)

STAT: High atomic number
High X-ray attenuation.



POWER: THE VESSEL-PAINTER.
Iodine turns blood into White-Flowing-Lava on a CT scan.

LORE: Finds Hank-the-Hemorrhage or Clot-the-Blocker. Caveat: Watch out for the kidneys!

NEPHRO-THE-REACTION (Contrast Nephropathy)

STAT: Creatinine spike
24-48h post-contrast.



POWER: THE KIDNEY-STUNNER.
Nephro strikes when Iodine-the-Ion is too harsh on the Renal-Tubes.

LORE: Prevented by Fluid-Flooding before the scan. High-risk for Diabetes-Dan.

Stay fluffy! Contrast tastes like HUMBUG!

21

SCOTTIE-THE-DOG
(Spondylolysis)STAT:
Broken Neck on
oblique lumbar X-ray.

POWER: The Spine-Snap. Shows a stress-fracture in the Pars-interarticularis.
LORE: Common in Gymnast-and-Wrestler athletes who arch their backs too much.

23

STEP-OFF-THE-JOINT
(Subluxation)STAT:
Malalignment of
articular surfaces.

POWER: The Staircase-Sign. Shows exactly where a joint has Slipped-the-Gears.
LORE: A stark indicator of Ligament-Laceration or unstable, sliding fractures.

22

FAT-PAD-FRED
(Elbow Fracture)STAT:
Sail Sign (Anterior) /
Posterior fat pad.

POWER: The Hidden-Blood-Detector. Shows the Shadow-of-Blood even if the fracture is invisible.
LORE: Blood filling the joint capsule physically pushes the normal fat pads out of their hiding spots.

24

SUNBURST-THE-SARCOMA
(Bone Tumor)STAT:
Spiculated
periosteal reaction.

POWER: The Bone-Explosion. Malignant Mas-Tumor grows so fast it blasts through the bone-armor.
LORE: The bone tries to build new periosteum armor but fails, leaving radiating Sunray-Spikes.

25 SHADOW-THE-STONE (Acoustic Shadowing)

STAT:
Black area deep to a
hyperechoic structure.



POWER: THE GALLSTONE-GUARD.
Cathy-the-Gallstone is so hard that
zero sound waves can pass through.

LORE: Sound is reflected 100% by
the surface, leaving the area behind
it absolutely Silent-and-Black.

27 DOPPLER-THE-DRIFTER (Color Flow)

STAT:
Frquency shift / Red
(Toward), Blue (Away).



POWER: THE FLOW-METER. Shows
if blood is blasting through the
Pipe-Highway or stalled by a clot.

LORE: Remember BART: Blue Away,
Red Toward. Essential for tracking
down Doris-the-DVT.

26 COMET-THE-TAIL (Ultrasound Artifact)

STAT:
Vertical reverberation
artifact.



POWER: THE METAL-FINDER.
Appears when sound hits a rigid
Needle or a Stent.

LORE: Sound bounces back and forth
infinitely inside the metal, confusing
the machine into seeing a long Tail.

28 SLUDGE-THE-SEDIMENT (Gallbladder)

STAT:
Low-level internal echoes
without shadowing.



POWER: THE STAGNANT-POOL.
Sludge is the Pre-Stone-Thickening
of bile that slowly shifts with gravity.

LORE: Micro-crystals and heavy
mucus settling quietly at the absolute
bottom of the Biliary-Bucket.

BART!
KayfaBONGO!



BART!
KayfaBONGO!



HOT-SPOT-THE-ISOTOPE

(Bone Scan)

STAT: Tc-90m MDP /
High osteoblastic activity.

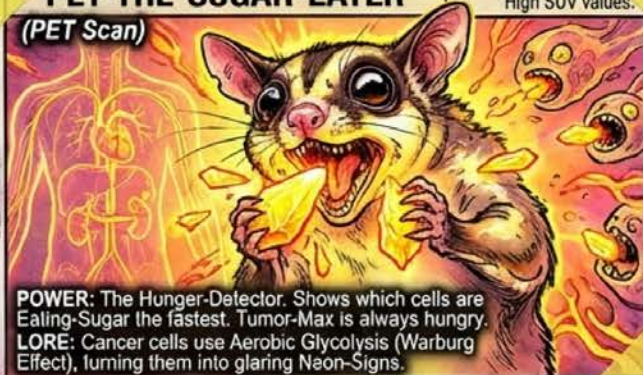


POWER: The Repair-Radar. Glows bright wherever Bone-Builders are working overtime to fix damage.
LORE: Excellent for sniffing out Mitch-the-Metastasis long before he shows up on a standard Roentgen-Ray.

PET-THE-SUGAR-EATER

(PET Scan)

STAT: FDG uptake /
High SUV values.



POWER: The Hunger-Detector. Shows which cells are Eating-Sugar the fastest. Tumor-Max is always hungry.
LORE: Cancer cells use Aerobic Glycolysis (Warburg Effect), turning them into glaring Neon-Signs.

COLD-NODE-THE-VOID

(Thyroid Scan)

STAT: Area of decreased
uptake on Iodine-123 scan.



POWER: The Cancer-Crate. A Cold-Node isn't making hormone. In ENT, this is highly suspicious.
LORE: Hot-Nodes are overactive but benign. Cold-Nodes are silent, sneaky, and potentially lethal.

V/Q-THE-MISMATCH

(Lung Scan)

STAT: Ventilation vs.
Perfusion imaging.



POWER: The Clot-Hunter. Finds an area of lung that is Breathing fine but has No-Blood-Flow.
LORE: The classic, unmistakable pathognomonic signature of a massive Pulmonary Embolism.

33 Seldinger-the-Entry

(Technique)

POWER: The Wire-Master. Seldinger lets the surgeon enter any Pipe in the body through a tiny pinhole.

LORE: The foundation of IR. Once the Guidewire-Ghost is inside, the Catheter-Cavalry can safely follow.

STAT: Needle → Wire → Catheter.



35 Stent-the-Scaffold

(Pipe Support)

POWER: The Pipe-Prop. Pushes Athero-Al's gunk tightly against the wall to keep the Highway open.

LORE: Installs permanent, unyielding structural support for a failing, narrowed Vessel-Wall.

STAT: Self-expanding vs. Balloon-expandable.



34 Embolo-the-Plug

(Embolization)

POWER: The Bleed-Stopper. Intentionally blocks a Leaky-Pipe to stop a hemorrhage or starve a tumor.

LORE: Creates a perfectly timed Strategic-Clot exactly where the plumbing is failing most.

STAT: Coils, Gelfoam, or Beads.



36 Drainage-the-Drain

(Abscess Care)

POWER: The Pus-Vacuum. Bypasses the Blade-Brigade to safely suck out infection through a tiny tube.

LORE: Relies on Gravity-and-Negative-Pressure to entirely drain the deeply buried Septic-Swamp.

STAT: Percutaneous pigtail catheter.



KayfaBIZARRO!
Plumb the depths!



37 FAST-Fiona (Trauma Sonography)

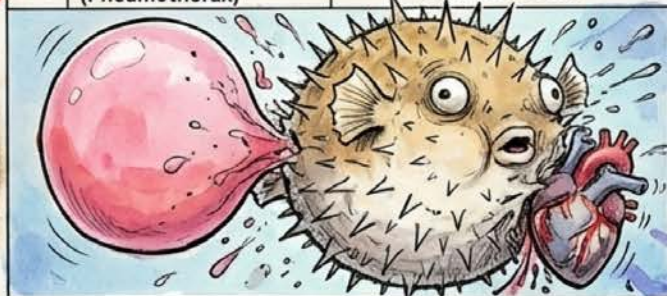
STAT: Focused Assessment with Sonography for Trauma.



POWER: The Blood-Seeker. Scans the belly and heart-sac in seconds to hunt for hidden hemorrhage.
LORE: A positive FAST-Fiona means the wrestler is actively bleeding internally—straight to the OR!

38 Tension-Tony (Pneumothorax)

STAT: Mediastinal shift away from the burst.



POWER: The Heart-Crusher. A one-way air valve inflates until it squashes the heart from pumping.
LORE: Primarily a clinical diagnosis, but Roentgen-Ray confirms it visually with a massive Vivid-Air-Space.

39 Deep-Sulcus-the-Sign (Hidden Air)

STAT: Abnormally deep costophrenic angle on supine X-ray.



POWER: The Hidden-Air-Detector. When a wrestler is lying flat, Tension-Tony's air hides at the bottom.
LORE: A subtle but incredibly lethal sign of a massive Pneumothorax in the supine ICU patient.

40 Aortic-the-Dissection (CT Angio)

STAT: Intimal flap / True vs. False Lumen



POWER: The Double-Pipe. Shows the Wall-Tear that splits the main aorta into two competing tunnels.
LORE: High pressure violently rips the layers apart. If the tear hits the Heart-Gate, it's game over.

41

Birads-the-Score
(Breast Imaging)STAT: Score 0 (Incomplete)
to 6 (Proven Cancer).

POWER: The Risk-Meter. Tells the surgeon exactly how worried they need to be about a Lump-Max.
LORE: A score of 4 or 5 is a massive red flag meaning Biopsy-Now!

43 Micro-Calc-the-Dots
(DCIS Indicator)STAT: Pleomorphic, branching
microcalcifications.

POWER: The DCIS-Signal.
 Tiny, jagged white dots
 that look like Cracked-Salt,
 often the only early sign.

LORE: Microscopic
 necrotic debris inside
 the breast ducts
 that has died and
 turned to stone.



MedKayfab!
 Look closely!



42

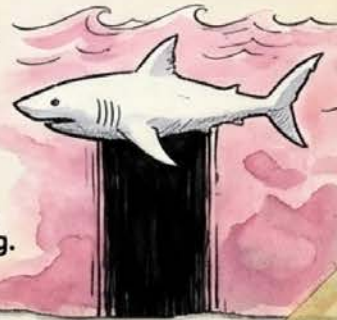
Spiculated-the-Shadow
(Mammography)STAT: Star-shaped mass
with radiating lines.

POWER: The Malignant-Signature. The distinct
 radiographic look of cancer Reaching-Out to invade.
LORE: Desmoplastic reaction physically shortens and
 pulls on the surrounding Cooper's-Ligaments.

44 Shadow-the-Acoustic
(Breast US)STAT: Posterior shadowing
on ultrasound.

POWER: The Solid-Finder.
 If the sound waves can't
 pierce through the lump,
 it's highly suspicious.

LORE: Malignant tissue
 is Dense-and-Absorbing,
 while benign simple cysts
 are Fluid-and-Transmitting.

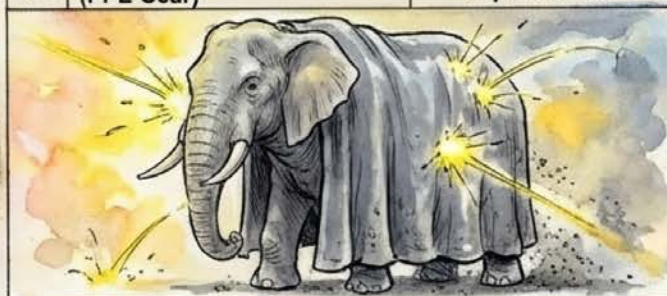


45

Alara-the-Protector
(Radiation Shield)STAT: As Low As
Reasonably Achievable.

POWER: The Radiation-Shield. Ensures we use the absolute minimum Roentgen-Rays needed to win.
LORE: Dictates Time, Distance, and Shielding. Use Echo or Mag whenever possible instead.

46

Shielding-the-Lead
(PPE Gear)STAT: 0.5mm Lead
equivalence.

POWER: The Photon-Stop. Lead armor stops stray Roentgen-Rays from hitting the Ovarian-Bench.
LORE: The incredibly high atomic number of lead absorbs the incoming X-ray energy before it hits DNA.

47

Pregnancy-the-Red-Flag
(Contraindication)STAT: High teratogenesis
risk in 1st trimester.

POWER: The Fetal-Defense. The supreme rule: always ask if there is a Fetal-Fighter in the fortress.

LORE: Ionizing radiation is infinitely more dangerous during the delicate Organ-Creation phase.



48

Metal-the-Missile
(MRI Hazard)STAT: Ferromagnetic objects
in the 3T field.

POWER: The Magnetic-Attraction. Mag-the-Magnet turns wheelchairs and pacemakers into Lethal-Projectiles.

LORE: The magnet is ALWAYS ON. It will rip ferromagnetic metal through the air like a fired rocket.



What the FLUFF?!
ALWAYS ON!



49

The PACS-the-Archive (Digital Library)

STAT: DICOM (Digital Imaging
and Communications in Medicine).



POWER: The Infinite-Memory. PACS stores every scan the wrestler has ever had for Old-vs-New battles.
LORE: Comparison is the key to Radiology. A terrifying lump from 2020 is much safer if it hasn't grown.

51 The Honey-Combing (Lungs)

STAT: Subpleural cysts
stacked in rigid rows.

POWER: The End-Stage-Lung. Chronic inflammation turns the flexible breathing balloons into useless cysts.

LORE: The irreversible sign of Pulmonary Fibrosis. The lungs are now permanent, unyielding stone.



KayfaBOGGLE!
Check the old
films!

50

The Apple-Core Sign (Colorectal)

STAT: Circumferential
narrowing of the colon.



POWER: The Cancer-Collar. Shows precisely where Malignant-Max has wrapped entirely around the colon.
LORE: The classic, definitive hallmark of advanced colon cancer visualized on a Barium-Enema.

52 The Thumb-Print Sign (Epiglottitis)

STAT: Swollen epiglottis
on lateral neck X-ray.

POWER: The Airway-Alarm. A massive warning that the wrestler's primary breathing tube is swelling shut.

LORE: The delicate epiglottis leaf swells up so large it looks exactly like a giant thumb on the X-ray.



KayfaBrite!
Check the
old films!



The Radiology Attending (Doc FrizzleBob)

STAT: Board Certified / Dark-Room Dweller.

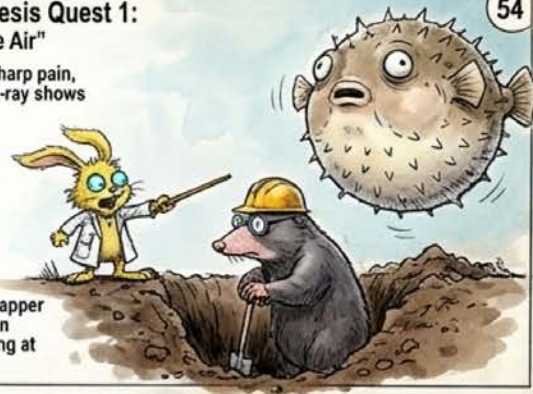


POWER: The Pattern-Recognizer.
LORE: Don't just look, see! Is the silhouette lost? Is the HU correct? Compare it to the old films or stay in the dark! Stay fluffy!

53

FrizzleBob's Synthesis Quest 1: Subtitle: "The Invisible Air"

LORE: Trauma wrestler, sharp pain, low SpO2. Supine chest X-ray shows Deep-Sulcus-Sign.



QUEST: Identify the Air-Trapper (Tension-Tony) and explain exactly why the air is hiding at the bottom of the ribs.

54

FrizzleBob's Synthesis Quest 2: Subtitle: "The Banana and the Lemon"

LORE: A CT head scan shows Crescent-Chris on the left side and Lemon-Lenny on the right side.



QUEST: Identify which bleed is the fast Arterial-Burst (Lenny) and which is the slow Venous-Leak (Chris).

55

FrizzleBob's Synthesis Quest 3: Subtitle: "The Golden Glow"

LORE: A PET-scan reveals an intense Sugar-Eating spot in the lung and another identical spot in the liver.



QUEST: Identify Malignant-Max and Metastasis-Mitch, and explain exactly why their cells glow so bright on the scan.

56

HUMBUO!
You did it!
Always
check the
old films!

HUMBUO!
You did it!
Always
check the
old films!

STAY FLUFFY!